

Tran Duc Dang Khoi

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Career Objective

- I am passionate about innovation and leveraging my experience across technology, communication, and research to drive outstanding product development.
- In the short term, my goal is to sharpen my skills and deepen my professional expertise in analytics, research, data science, artificial intelligence, and product development process optimization.
- In the long term, I aspire to contribute meaningful value to transformative product development initiatives and to participate in management and cross-functional collaboration within leading technology corporations.

Education

University of Engineering and Technology, VNU, Faculty of Artificial Intelligence 09/2022 – 7/2026

- GPA: 3.4/4.0
- Achieved grade A or above in Machine Learning, Deep Learning, Artificial Intelligence, Natural Language Processing, Object-Oriented Programming, and other core courses.

Work Experience

Mathematics Tutor, Ket Noi Gia Su Tutoring Center 10/2023 – Present

- Provided one-on-one mathematics tutoring to students from grades 8 through 12.
- Developed structured, personalized learning roadmaps ranging from foundational remediation to advanced preparation; multiple students achieved scores of 9.0 or above in national examinations.
- Reference: Ket Noi Gia Su Center – 0963672299

Makoto Trading and Services Joint Stock Company, Operations Manager 01/2024 – 12/2025

- Managed a team of 10 staff to maintain core business operations in the service and restaurant sectors.
- Reference: Mr. Ngo Quang Thanh – 0947142580

Extracurricular Activities

Deputy Head of Event Organization, SGUET – Student Support Club, University of Engineering and Technology 09/2023 – Present

- Organized large- and small-scale events including Warm Winter, BIGGAME, HOLA event series, Picnic, and Freshman Orientation Workshop, with attendance of up to several hundred participants.
- Assisted in organizing and supporting university admissions consulting activities for the University of Engineering and Technology.

Projects

Text-to-SQL System – Graduation Thesis github.com/KL

- Description: Developed a Text-to-SQL system translating natural language questions into executable SQL queries, applying NLP and large language model techniques.
- Role: Independently researched, designed, implemented, and evaluated the entire system as a solo graduation thesis project.
- Technologies: Python, NLP, LLM, SQL

PHhouse – Mini Apartment Management Website (Freelance) github.com/PHhouse

- Description: Built a web application for managing mini-apartment (CCMN) operations, including tenants, rooms, and payments, as a freelance client project.
- Role: Independently designed the database, developed the full-stack application, and deployed the system alone.
- Technologies: Full-stack Web Development

Educational Gamification – Intellius

github.com/Intellius

- Competition: IAI Hackathon 2023
- Description: Designed an educational game integrating AI to deliver interactive and adaptive learning experiences.
- Role: Conceptualized game mechanics and narrative, developed the user interface, and implemented core game logic.
- Technologies: Unity, JavaScript

AI-Based Cancer Diagnosis System

github.com/AI

- Description: Developed a web application employing Deep Learning models for cancer classification and segmentation from medical imaging data.
- Role: Collected, processed, and cleaned datasets; designed and trained the CNN model; conducted web application testing.
- Technologies: Python, CSS, SQLite3

Clara – Female Psychology Exploration Chatbot

github.com/chatbot

- Description: Built a chatbot designed to foster understanding of female psychology, utilizing Corrective RAG combined with Elasticsearch document retrieval based on Sparse Vector Embedding.
- Role: Collected, processed, and cleaned data; developed the chatbot interface; designed and integrated a large language model pipeline for data processing.
- Technologies: Python, TSX, LangChain, Elasticsearch

Mailshake – Secure Email System

github.com/secureMail

- Competition: Onchain Names and Identity Vietnam Hackathon
- Description: Built a secure email system featuring end-to-end encryption combined with blockchain technology (Handshake domain) to protect message confidentiality.
- Role: Developed smart contracts to record encrypted email content on the blockchain using Ethereum.
- Technologies: Solidity, MetaMask, Node.js, HTML, JavaScript

Image Captioning

github.com/image

- Description: Developed a system to generate descriptive captions for images, capturing objects, actions, and contextual relationships within scenes. Multiple deep learning architectures were implemented and benchmarked to evaluate their effectiveness in this domain.
- Role: Researched and integrated an LSTM-based deep learning model into the application.
- Technologies: Python

AI-Powered Movie Rental E-Commerce Website

github.com/Web

- Description: Developed a movie rental platform enabling users to stream content online, with an integrated AI recommendation engine to suggest films tailored to individual preferences.
- Role: Gathered, analyzed, and processed data; designed backend logic and database architecture; contributed to frontend development.
- Technologies: Python, Node.js, React, MongoDB, Express, TMDb API

Technical Skills

Programming Languages: C++ , Node.js, Python

Tools & Platforms: Docker, Git

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, LangChain

Foreign Languages: English (B2 – APTIS)

Achievements & Awards

IAI Hackathon 2023

- Finalist, IAI Hackathon 2023

Onchain Names and Identity Vietnam Hackathon 2024

- Top 10 Finalist, Onchain Names and Identity Vietnam Hackathon 2024

DEFONTIER 2024: “Notation”

- Top 20, Digital Economy Talent Search Competition 2024 – DEFONTIER 2024: “Notation”

Provincial Mathematics Excellence Award

- Second Prize, Quang Ninh Provincial Mathematics Gifted Student Competition

Illuminate the Future Scholarship

- Recipient of the Illuminate the Future Scholarship – DELOITTE

Outstanding Student Award

- Outstanding Student of the Academic Year 2023–2024, University of Engineering and Technology, VNU